



SIMATS
ENGINEERING



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(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Factory Auto Chain

GUIDED BY
DR. RamKumar M

BY
SWETHA K(192311404)
AYYAPPAN N(192411255)
PRADEEP P(192525367)



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Game Concept Overview

- A grid-based automation game inspired by Factorio.
- Players place machines (Extractor, Processor, Combiner) on a factory grid.
- Resources automatically move through connected machines.
- The goal is to design an efficient production chain that produces complex products.
- Win: Produce the target product with maximum efficiency.
- Lose: Incorrect machine connections or inefficient layouts reduce production.



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Reference Game Study

- **Reference Game:** Factorio

Why Factorio is Popular

- Automatic factory building and optimization.
- Resource extraction and production chains.
- Strategic planning and logistics.
- Continuous automation with minimal manual work.



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Our Simplifications

- 2D grid-based factory.
- Basic machine placement.
- Simple resource movement.
- No conveyor belt physics.
- Limited machine types.
- Simple recipes instead of hundreds



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Objectives

- Grid-based factory simulation.
- Java OOP concepts.
- Collections Framework for recipe management.
- Multithreading for automatic resource movement.
- Swing GUI for machine placement.
- JDBC with MySQL for production statistics.
- Exception handling for game reliability.



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Proposed Module Breakdown

Module	Description	Java Concepts Used
Module 1: Factory Grid & Machine Management	Creates the 2D factory grid, allows players to place Extractor, Processor, and Combiner machines, and manages machine positions and interactions.	OOP, 2D Arrays, Swing/AWT
Module 2: Resource Flow & Production Engine	Simulates automatic resource movement between machines, processes recipes, and produces final products using a background thread.	Multithreading, Collections (HashMap, ArrayList)
Module 3: Production Statistics & Database Management	Calculates production efficiency, tracks total resources produced, and stores player statistics in the MySQL database using JDBC.	JDBC, MySQL, Exception Handling

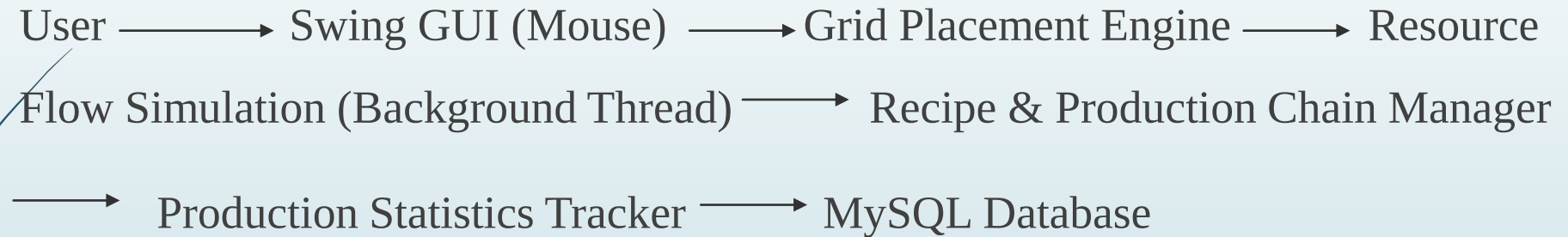


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Proposed Module Breakdown





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Tech Stack & Why

Technology	Usage
Java	Core programming language
Swing/AWT	Factory grid interface
OOP	Machine and resource classes
Collections	Recipe storage using HashMap and ArrayList
Multithreading	Continuous resource simulation
Exception Handling	Handle invalid operations and database errors
JDBC + MySQL	Store production statistics



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Proposed Database Schema

Column	Data Type
id	INT (Primary Key)
player_name	VARCHAR(50)
total_resources	INT
final_product	VARCHAR(50)
efficiency	DOUBLE
play_time	INT
date_played	DATE



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Multithreading Plan

Background Thread

- Runs continuously every second to:
- Move resources between connected machines.
- Process machine recipes.
- Update production count.
- Refresh factory status.

Why Multithreading?

- Without multithreading:
- GUI freezes.
- Resource movement stops.
- Factory animation becomes unresponsive.
- Poor user experience.



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Exception Handling Plan

Scenario	Exception	Solution
Machine placed on occupied cell	IllegalArgumentException	Show warning message
Invalid machine connection	NullPointerException	Ignore invalid connection
Database connection failure	SQLException	Display database error
Invalid recipe	NoSuchElementException	Skip processing
Empty machine inventory	IndexOutOfBoundsException	Prevent resource movement



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Work Plan

Week 1

- Design factory grid
- Create machine classes
- Develop GUI

Week 2

- Machine placement
- Recipe management
- Resource processing

Week 3

- Multithreading
- Production simulation
- Efficiency calculation

Week 4

- JDBC integration
- Database storage
- Testing
- Bug fixing
- Final presentation



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Work Plan

Challenge 1

Maintaining smooth resource movement between multiple machines.

Solution

- Use multithreading with synchronized resource updates.

Challenge 2

Managing complex production recipes.

Solution

- Store recipes using HashMap for fast lookup.



Engineer to Excel

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THANK YOU